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RESEARCH-ARTICLE

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Research on the Design of Digital Museum Information Presentation Based on Cognitive Load Theory

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Abstract

With the rapid development of digital technology, digital museums are increasingly attracting attention as important cultural and educational platforms. However, how to effectively design the information presentation of digital museums to optimise visitors' learning experience is still an urgent problem. Based on the cognitive load theory, this study explored the mechanism of the impact of the information presentation design of digital museums on visitors' learning effects. The study collected 308 valid samples through questionnaires, and used structural equation modelling to analyse the complex relationship between intrinsic cognitive load, extrinsic cognitive load, cognitive processing and learning effects. This study not only enriches the application of cognitive load theory in digital learning environments, but also provides an important theoretical basis and practical guidance for the design and optimisation of digital museums. The research results can help to enhance the effectiveness of digital museums as educational platforms and promote cultural learning and lifelong education for the public. Future research can further explore the cognitive load management strategies for different types of users, as well as the impact of emerging technologies such as AR/VR in digital museums on cognitive load and learning outcomes.

CCS Concepts

• **Information systems** → Information systems applications; Multimedia information systems; Multimedia databases.

Keywords

Digital museums, cognitive load theory;,information presentation design, learning outcomes

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1 INTRODUCTION

In the 21st century with the rapid development of digital technology, museums, as an important carrier of human civilisation, are undergoing unprecedented changes. The rise of digital museums not only breaks the time and space limitations of traditional museums, but also provides the public with richer, more interactive and personalised learning experiences. This transformation is both an opportunity and a challenge, especially in terms of information presentation and user learning experience, which raises a number of issues worthy of in-depth discussion. In recent years, digital museums have shown rapid growth. According to the International Council of Museums (ICOM), the number of digitised museums worldwide has grown from less than 1,000 in 2010 to more than 15,000 by 2023, with an average annual growth rate of more than 25%. This growth trend reflects the increasing importance of digitisation in the field of cultural communication and education. At the same time, visits to digitised museums are also showing significant growth. According to statistics, the average number of monthly visits to digital museums around the world reached 250 million in 2023, an increase of 150 per cent over 2019. This figure reflects the huge public demand for digitised cultural content.

However, growth in quantity does not equate to an increase in quality. According to a survey of 1,000 major digital museums by Museum Analytics: the average time spent on a digital museum website is 14 minutes per visitor. The percentage of visitors who engage deeply with content (e.g., watch a video in its entirety or participate in an interactive display) is only 35%. The repeat visit rate (multiple visits within a month) is only 22 per cent. The share of mobile visits has reached 65%, but only 40% of digital museums are optimised for mobile. This phenomenon raises the question of what prevents visitors from learning and exploring the content of digital museums more deeply?

Cognitive load theory provides an important theoretical perspective to answer this question. Proposed by John Sweller in 1988, the theory states that human working memory capacity is limited and that too much or inappropriate presentation of information may lead to cognitive overload, thus affecting learning [26]. In a digital museum environment, visitors often need to process multiple forms of information such as text, images, audio and video simultaneously, which undoubtedly puts great pressure on their cognitive systems.

Cognitive load theory divides cognitive load into three categories: intrinsic cognitive load, extrinsic cognitive load and related cognitive load. Intrinsic cognitive load in the context of digital museums relates to the complexity of the learning content itself, such as the causal relationship of historical events or the stylistic features of artworks. Extrinsic cognitive load relates to the way the information is presented, such as interface design, navigation structure and the use of multimedia elements. Relevant cognitive load refers to the cognitive processing that the learner undertakes in order to understand and remember the information [23].

Based on this theoretical framework, the information presentation design of digital museums can be better understood and optimised to reduce unnecessary cognitive load and improve learning outcomes. This study used a combination of questionnaires and structural equation modelling analysis. A questionnaire containing four dimensions, namely intrinsic cognitive load, extrinsic cognitive load, cognitive processing and learning effect, was designed and surveyed among visitors of several digital museums. Through structural equation modelling, the complex relationship between these factors will be analysed.

In summary, this study is based on the new challenges of museum education in the digital era and uses cognitive load theory to explore the impact of information presentation design on learning effects [24]. Through a combination of theoretical analysis and empirical research, we expect to provide valuable theoretical guidance and practical suggestions for the development of digital museums, and ultimately promote the public's better access to knowledge, appreciation of art, and understanding of history through digital platforms, so as to realise the digital transformation and enhancement of the museum's educational function.

2 LITERATURE REVIEW

2.1 Cognitive Load Theory

Cognitive Load Theory [1] (CLT) was developed by Australian educational psychologist John Sweller in the 1980s. This theory focuses on how the architecture of the human cognitive system and the limitations of working memory affect the learning process (Sweller, 1988). Cognitive load theory provides an important theoretical framework for understanding the mechanisms of information processing in the learning process, and is particularly valuable for application in complex learning environments. [2]

2.1.1 Concepts of Intrinsic Cognitive Load, Extrinsic Cognitive Load and Related Cognitive Load. Cognitive load theory classifies cognitive load into three types: intrinsic cognitive load, extrinsic cognitive load, and related cognitive load.

Intrinsic cognitive load is determined by the inherent complexity of the learning material. It depends on the difficulty of the learning task and the learner's prior knowledge (Sweller et al., 1998) [3]. For example, understanding the characteristics of Renaissance art may generate a high intrinsic cognitive load for a beginner, while the intrinsic cognitive load of this task is relatively low for an art history expert [4]. Chinese scholars Zhang Yi and Liu Rude (2006) point out that intrinsic cognitive load is unavoidable but can be managed through sound instructional design [5]. They suggest breaking down complex learning tasks into smaller, manageable units to reduce the intrinsic cognitive load at each learning stage [25].

Extrinsic cognitive load is caused by the way information is presented and the instructional design, which does not directly contribute to learning (Chandler & Sweller, 1991). For example, in a digital museum, a confusing layout of exhibit information or a complex navigation system can increase extrinsic cognitive load. Li Mang et al [6]. (2010) showed that in multimedia learning environments, rationally designing the interface and controlling the way information is presented can effectively reduce the extrinsic cognitive load and thus improve learning.

Relevant cognitive load refers to the cognitive resources required by learners in the process of constructing and automating schemas (Sweller et al., 1998). This load is beneficial because it directly promotes learning. Domestic scholars Peng Shaodong and Huang Shan (2009) emphasised that instructional design should aim to reduce the extrinsic cognitive load and at the same time increase the relevant cognitive load in order to promote deep learning and knowledge transfer [7].

2.1.2 Cognitive Load Theory in Education and Learning. Cognitive load theory has been widely used in education and learning, especially in complex learning environments and multimedia learning [8].Mayer and Moreno (2003) proposed a series of multimedia learning principles based on cognitive load theory, such as the principle of separation and the principle of coherence, which are aimed at reducing the external cognitive load and optimising the learning effect. Domestic scholars Zhao and Gu (2011) applied the cognitive load theory to online course design and proposed strategies including information chunking and progressive presentation, which effectively reduce the cognitive load of learners.Kalyuga (2007) proposed an adaptive teaching system based on the cognitive load theory, which dynamically adjusts the complexity of the learning content according to the learner's level of expertise to optimise the intrinsic cognitive load. Zhang Lixin et al. (2015) developed an intelligent tutoring system on this basis, which dynamically adjusts the difficulty and degree of support of learning tasks by monitoring the cognitive load status of the learners in real time.Andersen et al [20]. (2016) investigated the problem of cognitive load in virtual reality environments, and found that an appropriate interaction design can reduce the extrinsic cognitive load and improve learning engagement [14]. Domestic scholars Liu Dejian et al. (2019) applied cognitive load theory in augmented reality teaching and proposed the design principle of "cognitive load balance" to ensure that technological enhancement does not interfere with the learning process. Through the above applications, cognitive load theory not only provides a new perspective for traditional teaching, but also provides important theoretical guidance for the application of emerging educational technology [15].

2.2 Concepts of Intrinsic Cognitive Load, Extrinsic Cognitive Load and Related Cognitive Load

The design of information presentation in digital museums is the key to ensure their effective operation and fulfilment of educational

functions. In recent years, research in this area has focused on the following areas.

Marty (2008) emphasised the importance of the intuitiveness and ease of use of the user interface for the digital museum access experience [18]. He proposed principles of interface design based on user needs, including a clear navigation system and a consistent visual style. Based on user experience theory, domestic scholars Chen Jing and Wang Yan (2016) proposed the "4E model" of interface design for digital museums, i.e., Easy to use, Esthetics, Educational and Entertaining [30].

Styliani et al. (2009) investigated the application of 3D technology in digital museums and concluded that high-quality 3D models can provide a more realistic exhibit experience, but also pointed out the challenges of technical implementation. Zhang Yuxia and Li Yi (2018) explored the application of virtual reality (VR) technology in digital museums, and Skov and Ingwersen (2014) studied information retrieval behaviour in digital museums and found that users' retrieval strategies are often exploratory, which puts new requirements on information organisation in digital museums [13]. Domestic scholars Liu Wei and Zhang Chunjing (2020) proposed a knowledge graph-based information organisation method for digital museums, which improves the discoverability and utilisation of information through semantic associations [19].

Not and Petrelli (2019) developed a personalised recommendation system based on user interests and browsing behaviour, which can dynamically adjust the exhibit recommendation strategy and improve user satisfaction [9]. Nan Li et al. (2021) proposed a hybrid recommendation algorithm that combines content analysis and collaborative filtering, which increases the diversity of recommendations while ensuring their accuracy. These studies show that the design of information presentation in digital museums is moving towards being more user-friendly, technologically advanced and personalised. However, how to control the cognitive load and improve the learning effect in rich information presentation is still an issue that deserves in-depth research.

3 RESEARCH METHODOLOGY

This study used a cross-sectional survey design combined with structural equation modelling (SEM) to analyse the data in order to explore the impact of digital museum information presentation design on visitors' learning outcomes. Although this method cannot directly establish causality as in an experimental design, it is able to make reasonable inferences about the relationships between variables through rigorous theoretical derivation, multi-stage stratified random sampling, and path analysis with structural equation modelling. To enhance the internal validity of the study, a research model was constructed based on solid theoretical foundations to ensure the theoretical soundness of the relationships between variables. Multi-stage stratified random sampling method is used to improve the representativeness of the sample and reduce the selection bias [21]. Detailed reliability and validity analyses were conducted to ensure the reliability and validity of the measurement tools. Use structural equation modelling for path analysis, considering both direct and indirect effects to provide a more comprehensive picture of variable relationships. Robustness testing using the Bootstrap method enhances the credibility of the results [22].

3.1 Current Status of Digital Museum Research

Based on the variable definitions, a mediation model is constructed in which intrinsic and extrinsic cognitive loads are used as independent variables, cognitive processing as a mediator variable, and learning effects as a dependent variable. Feedback and optimisation can be used as complementary analyses.

HYPOTHESIS PROPOSED: The following hypotheses are proposed based on theory and previous research:

H1: Intrinsic cognitive load has a negative and significant effect on learning outcomes

H2: Extrinsic cognitive load has a negative and significant effect on learning outcomes.

H3: Cognitive processing mediates the relationship between cognitive load and learning outcomes.

3.2 Questionnaire Design

3.3 Data collection

According to Table seen from Table 1, In this study, a multi-stage stratified random sampling method was used to select the survey respondents in order to increase the representativeness and diversity of the sample. Firstly, the target population was divided into different age strata (18-30 years old, 31-45 years old, 46-60 years old) based on demographic data, and within each stratum, it was further subdivided by educational background (high school and below, college/undergraduate, postgraduate and above). Subsequently, we used systematic random sampling to select samples from each subdivided stratum, with sample sizes corresponding to the stratum's proportion of the overall population. To ensure occupational diversity, the proportion of different occupational groups (e.g., students, office workers, freelancers, educators, etc.) was monitored and adjusted during the sampling process to match the occupational distribution of the target population.

To verify the representativeness of the sample, the sample characteristics were compared with the latest census data to ensure that the sample proportions at each level did not differ significantly from the actual population distribution. In addition, a weighting adjustment technique was employed to further minimise possible sampling bias. A total of 350 questionnaires were distributed and 308 questionnaires were validly returned, representing an effective response rate of 88.0 per cent. In order to assess non-response bias, a short follow-up survey was conducted with non-respondents, which showed no significant difference in key characteristics between the non-response group and the response group. This optimised sampling and validation methodology aims to obtain a user group that accurately reflects different backgrounds and characteristics, thereby enhancing the generalisability and reliability of the findings. Through these rigorous sampling procedures and validation steps, the quality and representativeness of the data were improved, laying a solid foundation for subsequent analyses.

3.4 Analysis methods

3.4.1 Descriptive Statistical Analysis. Sample Characteristics The young group (18-34 years old) is the main user, occupying as much

Table 1: Model variable definitions and questionnaire reference scales.

Variant	Define	Question item	Resource
Intrinsic cognitive load	Personal characteristics such as prior knowledge, learning ability and interest in the subject matter of the interviewer	5-9	Paas et al. (2003)
External cognitive load	External features of digital museums such as information presentation, interaction design and technology application	10-14	Sweller et al. (1998)
Cognitive processing process	Visitors' mental processes for processing information in a digital museum environment	15-19	Mayer (2002)
learning effect	Visitors' ability to understand, remember and apply digital museum content	20-24	Bloom's Taxonomy of Educational Objectives (1956)
Feedback and optimization	Willingness and content to provide recommendations for improvements in digital museum design based on learning experiences	25-29	Self-designed with reference to UX research methodology

Note: The design of this questionnaire makes comprehensive reference to cognitive load theory, multimedia learning theory and user experience research methods, and has been appropriately adjusted according to the characteristics of digital museums.

Table 2: Exploratory factor analysis (EFA) results

Factor	Eigenvalue	Variance explained rate(%)	Cumulative variance explained(%)
Intrinsic cognitive load	4.327	21.635	l21.635
Extrinsic cognitive load	3.891	19.455	41.09
Cognitive processing	3.754	18.77	59.86
Learning effect	3.298	16.49	76.35

as 66.89%, meaning that they not only represent the future consumption trend, but also lead the trend of the times [10]. The sample population shows a balanced distribution in terms of gender, demonstrating the diversity and tolerance of modern society. Meanwhile, the sample population generally possesses a high level of education, with 64.61% of them having a bachelor's degree or above, showing a strong desire for knowledge and excellent learning ability. In terms of occupational composition, students, IT/technical personnel and educators constitute the main force of the sample, each with their unique perspectives and expertise, shaping the appearance and future of this era.

3.4.2 Descriptive Statistical Analysis. The following reliability analysis is carried out first, and Cronbach's alpha coefficient is used for the reliability test. As shown in Table 2, The Cronbach's alpha coefficients for the four variables of intrinsic cognitive load, extrinsic cognitive load, cognitive processing, and learning effect are 0.876, 0.892, 0.901, and 0.885, respectively, which are all greater than 0.8, indicating that the questionnaire has good internal consistency.

Then KMO and Bartlett's spherical test were conducted, and the results showed that the KMO value was greater than 0.8, which indicated that the sample was very suitable for factor analysis; the significance of Bartlett's spherical test was less than 0.001, which indicated that there was a correlation between the variables, and it was suitable for factor analysis.

Exploratory factor analysis (EFA) successfully extracted four factors corresponding to intrinsic cognitive load, extrinsic cognitive load, cognitive processing and learning effect. Principal component analysis (PCA) was used to extract the male factors, and orthogonal rotation was performed using the maximum variance method. A total of four factors were extracted, which explained 76.35% of the total variance, indicating that the questionnaire was well structured.

As shown in Table 3. The structural validity of the questionnaire was further confirmed by the results of the validated factor analysis (CFA). All factor loadings were greater than 0.5 and statistically significant ($p < 0.001$). Parameter estimation was performed using the maximum likelihood method and multiple fit metrics were used to assess the fitness of the model.The model fit indices all reached the good level, indicating that the measurement model has good structural validity. All of the fit indicators reached the good level, indicating that the measurement model fits the actual data well.

3.4.3 Relevance Analysis. As shown in Table 4, The results of the analyses show that intrinsic cognitive load is moderately positively correlated with extrinsic cognitive load; this indicates that when learners perceive a higher intrinsic cognitive load, they also tend to perceive a higher extrinsic cognitive load and vice versa. This reflects a correlation between the complexity of the learning content (intrinsic load) and the complexity of the interface design (extrinsic load) in the digital museum environment. Both intrinsic and extrinsic cognitive load are negatively correlated with cognitive

Table 3: Exploratory factor analysis (EFA) results

Goodness-of-fit indicator	Numerical value	Criteria for judgement	In the end
χ^2/df	2.34	<3	Favourable
RMSEA	0.056	<0.08	Favourable
CFI	0.943	>0.9	Favourable
TLI	0.936	>0.9	Favourable

Table 4: Exploratory factor analysis (EFA) results

Variant	Intrinsic cognitive load	Extrinsic cognitive load	Cognitive processing	Learning effect
Intrinsic cognitive load	1			
Extrinsic cognitive load	0.321**	1		
Cognitive processing	-0.412**	-0.537**	1	
Learning effect	-0.385**	-0.492**	0.678**	1

*Note: ** $p < 0.01$*

Table 5: Path factor analysis

Trails	Standardised path factor	T-Value	P-Value
Intrinsic cognitive load → Cognitive processing	-0.323	-5.876	0.002
Extrinsic cognitive load → Cognitive processing	-0.462	-8.213	0.017
Cognitive processing → Learning effect	0.712	14.567	0.004
Intrinsic cognitive load → Learning effect	-0.134	-2.987	0.000
Extrinsic cognitive load → Learning effect	-0.187	-3.654	0.002

processing and learning outcomes; this means that when learners perceive a higher intrinsic cognitive load, their cognitive processing may be hindered [11]. It reflects that the complexity of learning content affects learners' information processing ability. Cognitive processing shows a strong positive correlation with learning effectiveness. It shows that learners' cognitive processing is closely related to the final learning effect.

3.4.4 Structural Equation Modelling Analysis. Prior to the structural equation modelling analysis, the data were first tested for normality, multicollinearity and sample size adequacy. By looking at the skewness and kurtosis of the variables, it was confirmed that the data generally conformed to the assumption of normal distribution (absolute value of skewness < 3 and absolute value of kurtosis < 10 for all variables). The test for multicollinearity showed that the variance inflation factor (VIF) for all variables was less than 5, indicating that there was no serious problem of multicollinearity. The sample size (N=308) fulfils the recommendation of at least 10 samples for each estimated parameter.

According to the data in Table 5, it can be seen that the intrinsic cognitive load has a significant negative effect on the cognitive processing process, i.e., the complexity of the learning content will hinder the cognitive processing of the learners. Therefore, it is suggested that when designing digital museum content, learners' cognitive ability needs to be taken into account and the content difficulty needs to be adjusted appropriately.

Secondly, external cognitive load has a stronger negative effect on cognitive processing. It indicates that factors such as interface design and navigation of digital museums have a greater impact on learners' cognitive processing. The importance of optimising the user interface and improving usability is emphasised. Suggests that factors such as interface design and navigation of digital museums have a greater impact on learners' cognitive processing. The importance of optimising the user interface and improving usability is emphasised.

Cognitive processing has a strong positive influence on learning outcomes. It is shown that supporting and facilitating learners' cognitive processing is key to improving learning outcomes. Suggests designing interactive features and learning activities that promote deep cognitive processing [12].

Intrinsic cognitive load has a direct negative effect on learning effectiveness, although the effect is relatively small. It suggests that intrinsic cognitive load has a direct impact on learning outcomes even when the mediating role of cognitive processing is taken into account. The importance of reasonable control over the complexity of learning content is emphasised. Extrinsic cognitive load also has a direct negative effect on learning effectiveness, with a slightly larger effect than intrinsic cognitive load. It shows that the design factors of digital museums not only affect the cognitive processing process by influencing the cognitive processing, but also directly affect the learning effect. The importance of optimising user interface design is again emphasised.

Table 6: Table of intermediation effects(1)

Step	Trails	Ratio	Significance
Step1:independent variable→Implicit variable	Intrinsic cognitive load → Learning effect	-0.385	0.017
	Extrinsic cognitive load→ Learning effect	-0.492	0.025
Step2:independent variable→	Intrinsic cognitive load → cognitive processing	-0.412	0.006
intermediary variable	Extrinsic cognitive load → cognitive processing	-0.537	0.015
Step3:independent variable+	Intrinsic cognitive load → Learning effect	-0.134	0.002
intermediary variable→Implicit variable	Extrinsic cognitive load → Learning effect	-0.187	0.000
	Cognitive processing → Learning effect	0.712	0.041

Table 7: Table of intermediation effects(2)

Type of cognitive load	Airect ffect	Calculation of indirect effects	Indirect effect	Aggregate effect	Percentage of intermediary effects
Intrinsic cognitive load	-0.134	-0.412 * 0.712	-0.293	-0.427	68.62%
Extrinsic cognitive load	-0.187	-0.537 * 0.712	-0.382	-0.569	67.13%

Table 8: Table of intermediation effects(2)

Assume	Element	In the end	Clarification
H1	Intrinsic cognitive load has a negative and significant effect on learning outcomes	Support	Both direct (-0.134) and indirect (-0.293) effects were significant
H2	Extrinsic cognitive load has a negative and significant effect on learning outcomes	Support	Both direct (-0.187) and indirect (-0.382) effects were significant
H3	Cognitive processing mediates between cognitive load and learning outcomes	Support	Significant mediation effects, accounting for 67-69 per cent of the total effects

To further validate the stability of the model, a bootstrap (Bootstrap) analysis was conducted, using 1000 replicated samples. The results showed that the 95% confidence intervals of all path coefficients did not contain zero, further supporting the findings of this paper. In addition, the explained variance (R^2) of the model was calculated. The results showed that the model explained 43.7% of the variance in cognitive processing and 63.2% of the variance in learning effects, indicating that the model has good explanatory power.

3.4.5 Analysis of Intermediation Effects (Three-tep Approach). Cognitive processing played a significant partial mediating role between intrinsic and extrinsic cognitive load and learning effects, with the mediating effect accounting for more than 65% of the total effect in each case. According to the data in Table 6 And Table 7 that cognitive processing is an important mechanism connecting cognitive load and learning effects.

3.4.6 Hypothesis validation results. Table 8 shows the hypotheses and results of this study.

3.4.7 Hypothesis validation results. This study used a multi-stage stratified random sampling method and Structural Equation Modelling (SEM) analysis to investigate the mechanism of the impact of digital museum information presentation design on visitors' learning outcomes. The study sample (N = 308) was well represented and the questionnaire instrument demonstrated excellent reliability and validity (Cronbach's $\alpha > 0.8$, KMO = 0.892, $p < 0.001$). The results of the empirical analyses supported all the research hypotheses, revealing the following key findings.

From the perspective of cognitive load effects, both intrinsic cognitive load (β = -0.427, $p < 0.01$) and extrinsic cognitive load (β = -0.569, $p < 0.01$) had a significant negative effect on learning outcomes, with extrinsic cognitive load having a more prominent effect. The mediating effect inside the cognitive processing process plays a significant partial mediating role between cognitive load and learning effect, with the mediating effect accounting for 68.62% (intrinsic cognitive load) and 67.13% (extrinsic cognitive load) of the total effect, respectively. In terms of model fit, the structural equation model demonstrated goodness of fit (χ^2/df = 2.34, RMSEA = 0.056, CFI = 0.943, TLI = 0.936) and explanatory power (R^2 Cognitive Processing = 43.7%, R^2 Learning Effect = 63.2%).

4 CONCLUSION

(1) Optimising interface design to reduce the extrinsic cognitive load is a key strategy for enhancing learning outcomes in digital museums. Studies have shown that well-designed user interfaces can significantly reduce learners' cognitive load, thus enabling them to allocate more cognitive resources to the actual learning content. To achieve this goal, this study suggests the use of simple and intuitive interface layouts to minimise visual distractions. This

includes the use of consistent colour schemes and fonts, optimised page layout to highlight important information, and careful use of decorative elements [17]. At the same time, optimising the navigation system is essential for efficient information retrieval. It is recommended to implement clear navigation menus, breadcrumb navigation and powerful search functions, which not only help users locate exactly the information they need, but also enhance their understanding of the overall information architecture [27]. In addition, the judicious use of multimedia elements is an important consideration in balancing information richness and cognitive load. It is recommended that multimedia content such as images and videos be carefully selected and placed to ensure that they are closely related to the learning objectives, and that thumbnails and synopses of the multimedia content be provided so that users can independently decide whether or not to view the detailed content in depth. By implementing these design strategies, the external cognitive load of users can be effectively reduced, thus optimising the overall learning experience and effect [31].

(2) The design of personalised learning experience is important for the effective management of intrinsic cognitive load. The study points out that the match between the complexity of learning content and learners' prior knowledge directly affects the level of intrinsic cognitive load. Therefore, this study suggests implementing a multi-level content structure design to accommodate visitors with different knowledge backgrounds [28]. This approach allows users to choose the appropriate content depth according to their knowledge level and learning needs, ranging from simple descriptions to detailed analyses, effectively regulating the intrinsic cognitive load. Meanwhile, the introduction of adaptive learning systems is a powerful tool for optimising the personalised learning experience. By developing a SOPHISTICAL user profiling system, combined with machine learning algorithms, real-time content adjustment based on user behaviour can be achieved. This not only includes dynamic difficulty adjustment mechanism, but also involves the recommendation of personalised learning paths, thus optimising the user's learning sequence and process. In addition, the design of scientific interest assessment tools is crucial for improving learning motivation and engagement. It is recommended to develop concise and effective interest questionnaires and combine them with collaborative filtering algorithms to achieve accurate personalised content recommendation. Through the combined application of these strategies, the level of personalisation of the learning experience can be significantly enhanced and the intrinsic cognitive load can be effectively managed, thus optimising the overall learning effect [16].

(3) Promoting effective cognitive processing is the key link between cognitive load management and learning effect enhancement. Research has shown that in-depth cognitive processing plays a decisive role in the internalisation of knowledge and the formation of long-term memory. Therefore, this study suggests designing a series of guided questions and reflective exercises to deepen learning understanding [32]. This includes adding well-designed reflection questions after each exhibit or thematic unit to encourage users to think critically and reflect deeply [29]. At the same time, providing immediate feedback mechanisms can help users validate their understanding and promote depth and accuracy of cognitive processing. In addition, creating opportunities for knowledge integration and application is important for promoting learning transfer. It is recommended to develop interactive virtual labs, historical scene reenactments, and other features that enable users to apply what they have learnt in a simulated environment to deepen their understanding [33]. Introducing gamification elements is another strategy worth considering to facilitate knowledge transfer and application by adding fun to learning. Designing features such as point and badge systems, virtual tour guides, and knowledge competitions can not only increase user engagement, but also stimulate their intrinsic motivation to learn. Through the combined use of these methods, the deep cognitive processing of users can be effectively promoted, thus achieving optimal management of cognitive load and significant enhancement of learning effects.

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